

AP CSP | Chapter 6: Community Shade Map

Part A | Access, representation and evidence

Name: _____ Partner: _____ Date: _____

Purpose: collect residents' reports of shade needs and compare North/South priorities. Use Chapter 6 Impact Lab. Predict before revealing results; record model evidence, not claims about real residents.

1. Access model | Which barriers does your plan remove?

100 potential participants, in mutually exclusive groups: 40 with no barriers; 20 network only; 15 device only; 10 skills only; 5 accessible-interface only; 10 both network and device. All of a person's barriers must be removed for modeled access.

Support costs: network 2, device 2, skills 1, accessible interface 1. Use budget 4. Each support removes its named barrier for everyone who has it. Access does not guarantee participation.

Support plan / cost	Predicted people with access	Observed model result
No support / 0		
Network + device / ____		
Your plan: _____ Cost: ____		

Explain your plan using the group counts. State one real-world barrier or assumption that the model does not test.

2. Representation model | Sample mix and population mix

Population: North 60%, South 40%. Constructed need rates within groups: North 20%, South 80%. Keep those rates fixed. Weight the rates by the indicated mix; report percentages.

Calculation	Predicted rate / working	Observed rate
Sample mix 80% / 20%		
Population weights 60% / 40%		
Sample mix 60% / 40%		

What changes when the mix changes? Why can population weighting still miss bias within either group?

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Part B | Privacy, security, reuse and a recommendation

3. Data and protection | Use #security; do not enter real personal data.

Fields: zone, shadeRating, name, email, exactGPS. Goal: North/South priorities.

Decision	Choice and reason
Minimum fields to retain	
One field to omit	
Aggregate vs raw: benefit + residual risk	

Give two different authentication-factor categories, an example of each, and a risk they reduce.

Model only: whose public key encrypts a confidential message to Blake, and whose private key decrypts it?
Contrast symmetric encryption.

4. Reuse decisions | Apply the four simplified #responsibility cards.

Choose reuse with conditions, request permission, or replace. Cite the stated permission and required credit/change notes. Online access alone is not permission.

Card	Action / permission evidence / credit or change note
Shade icon	
Research figure	
Your original code	
Online photo	

5. Recommend a plan. Cite both pages; include who benefits, a tradeoff and an open question.